

CompTIA.

A+ Core 1

Exam 220-1201

A+ 220-1201

Module 4



Troubleshooting PC Hardware

Learning Outcomes

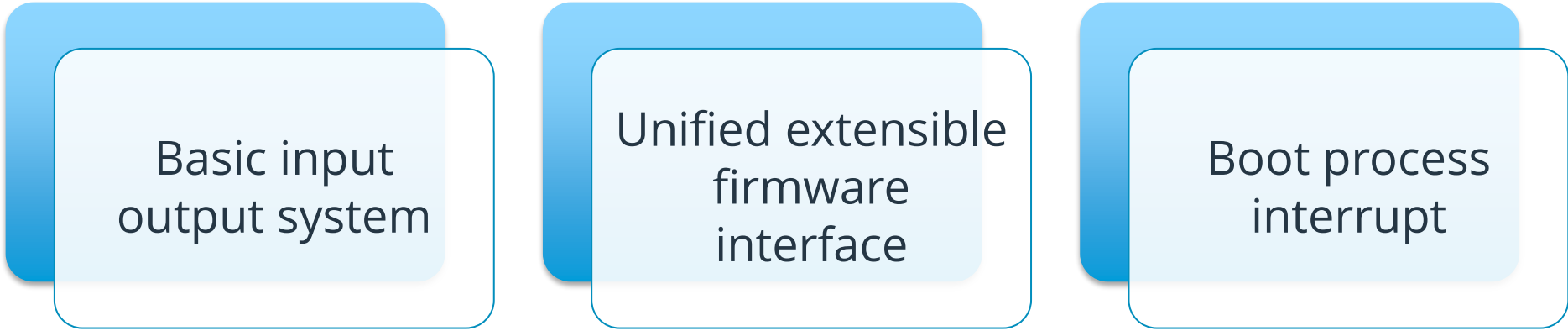
Prepare for A+ Core 1 by:

- Applying troubleshooting methodology
- Configuring BIOS/UEFI
- Troubleshooting power and disk issues
- Troubleshooting system and display issues

Lesson 4.1

BIOS and UEFI

BIOS and UEFI (Slide 1 of 4)



Basic input
output system

Unified extensible
firmware
interface

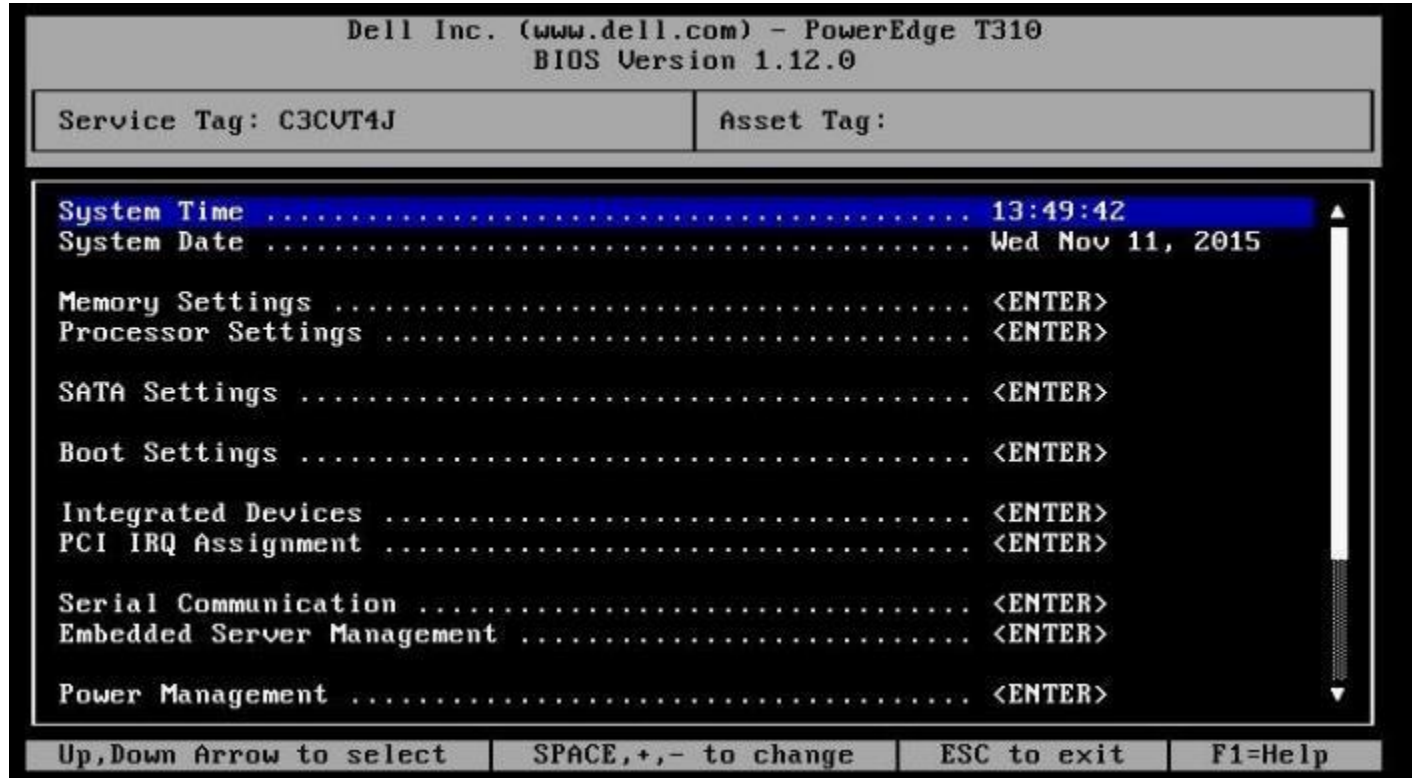
Boot process
interrupt

BIOS and UEFI (Slide 2 of 4)



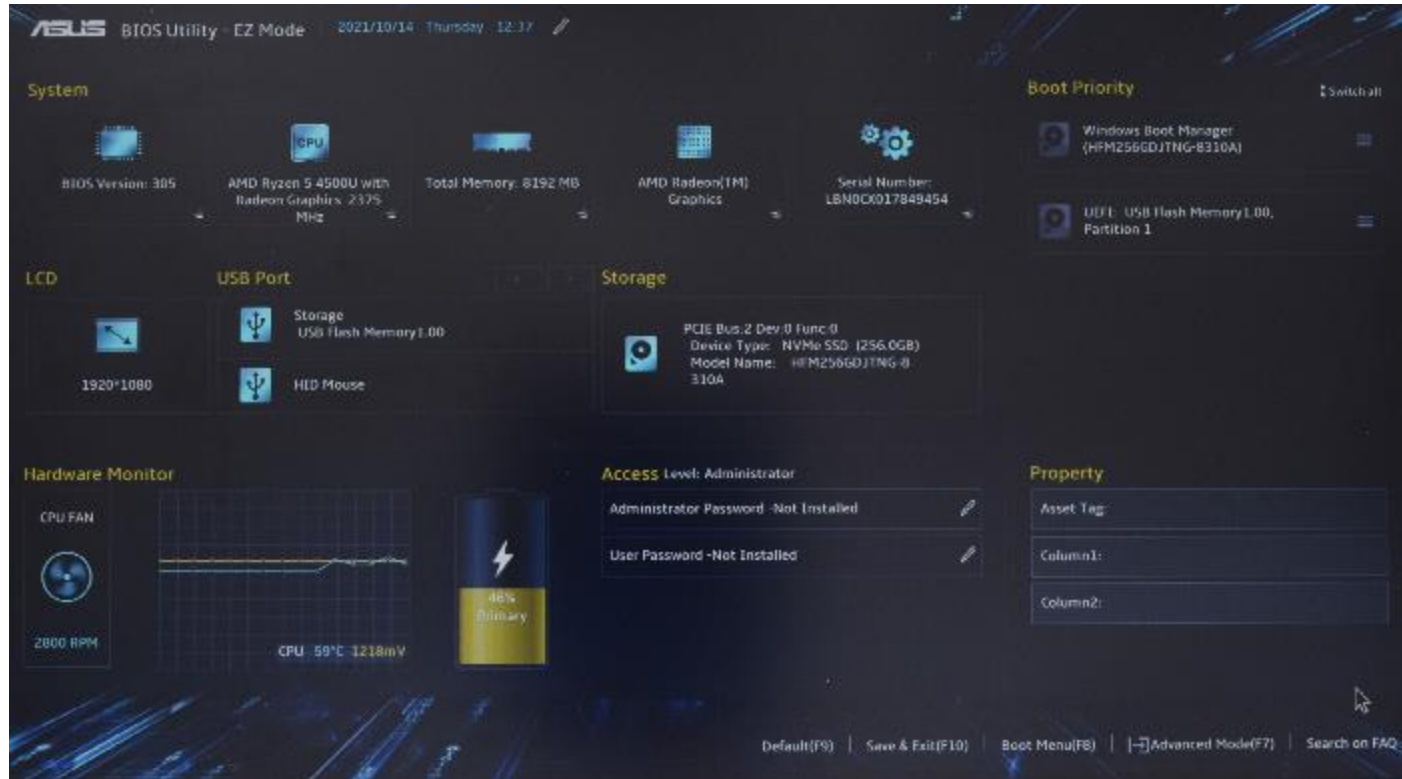
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BIOS and UEFI (Slide 3 of 4)



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BIOS and UEFI (Slide 4 of 4)



Screenshot used with permission from ASUSTek Computer Inc.

Boot and Device Options (Slide 1 of 3)

- Boot parameters
 - Fixed disk
 - Optical drive
 - USB
 - Network/PXE

Boot and Device Options (Slide 2 of 3)

Dell Inc. (www.dell.com) - PowerEdge T310
BIOS Version 1.12.0

Service Tag: C3CUT4J	Asset Tag:
----------------------	------------

SATA Settings <ENTER>

Boot S **Boot Mode BIOS**

 Boot Sequence <ENTER>

Integr Hard-Disk Drive Sequence <ENTER>

PCI IR USB Flash Drive Emulation Type <ENTER>

 Boot Sequence Retry Disabled

Serial

Embedded Server Management <ENTER>

Power Management <ENTER>

System Security <ENTER>

Keyboard NumLock On

Report Keyboard Errors Do Not Report

F1/F2 Prompt on Error Enabled

Up,Down Arrow to select SPACE,+, - to change ESC to exit F1=Help

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Boot and Device Options (Slide 3 of 3)

Dell Inc. (www.dell.com) - PowerEdge T310
BIOS Version 1.12.0

Service Tag: C3CUT4J	Asset Tag:
----------------------	------------

SATA S

Boot S

Integr

PCI IR

√ 1. SATA Optical Drive

2. Embedded NIC 1 BRCM MBA Slot 0200 v7.2.3

√ 3. Hard drive C: (Slot 1 #0300 ID4F LUN0 Dell Vi)

SPACE to enable/disable +,- to move down/up

Serial

Embedded Server Management <ENTER>

Power Management <ENTER>

System Security <ENTER>

Keyboard NumLock On

Report Keyboard Errors Do Not Report

F1/F2 Prompt on Error Enabled

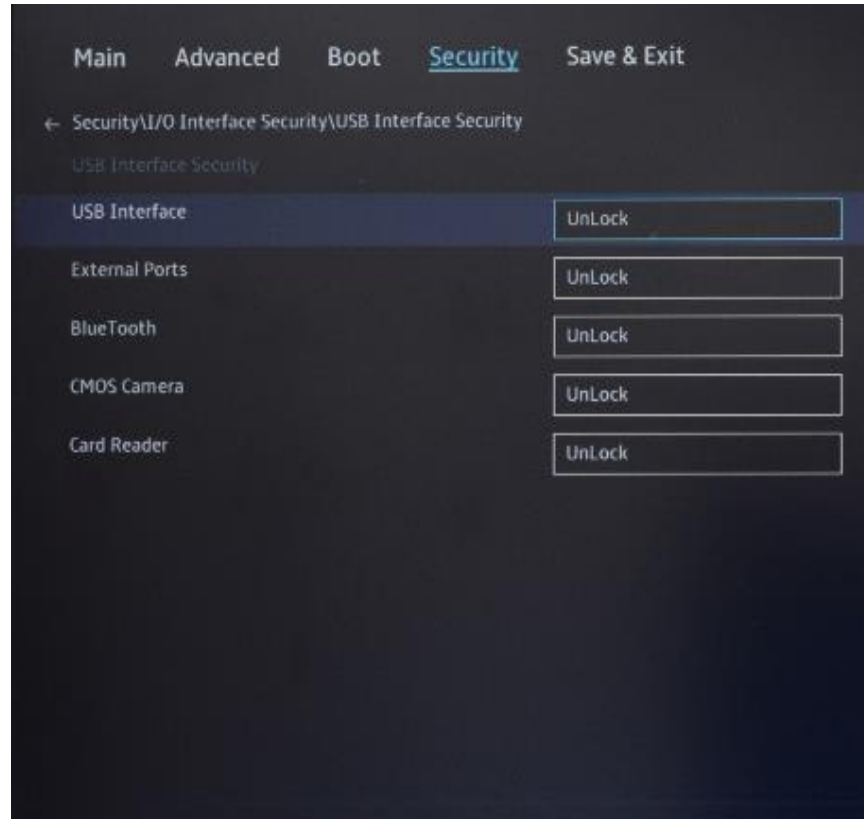
Up,Down Arrow to select SPACE,+,- to change ESC to exit F1=Help

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USB Permissions

- Enable and disable
 - USB controllers
 - Adapters
 - Ports
 - Individual
 - All

Boot and Device Options



Screenshot used with permission from ASUSTek Computer Inc.

Fan Considerations



Optimized performance



Balance power and speed



Temperature monitoring

Manual
Third-party application



Lesson Review: Boot Passwords and Secure Boot (Slide 1 of 2)



Supervisor/Admin/BIOS
password

Restrict access to make
changes in BIOS/UEFI menu



User system password

Locks system



Secure boot

Digital signature

Boot Passwords and Secure Boot (slide 2 of 2)



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Trusted Platform Modules (Slide 1 of 2)

- Securely store
 - Digital certificates such as those for secure boot
 - Cryptographic keys
 - Hashed passwords
- Hardware Security Module (HSM)
 - USB drive to store keys

Trusted Platform Modules (Slide 2 of 2)

Dell Inc. (www.dell.com) - PowerEdge T310
BIOS Version 1.2.1

Service Tag:	Asset Tag:
--------------	------------

SA	System Password Enabled
Bo	Setup Password Not Enabled
	Password Status Unlocked
In	
PC	TPM Security On with Pre-boot Measurements
	TPM Activation Activate
Se	TPM Clear No
Em	
	Power Button Enabled
Po	NMI Button Disabled
Sy	AC Power Recovery Last
	AC Power Recovery Delay Immediate
Ke	User Defined Delay <ENTER>
Re	
	F1/F2 Prompt on Error Enabled

Up,Down Arrow to select SPACE,+, - to change ESC to exit F1=Help

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Lab Activity

- Lab: Find BIOS/UEFI Settings
(available in CertMaster Learn and CertMaster Perform)
- Lab: Configure BIOS/UEFI Security
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: BIOS and UEFI

- BIOS and UEFI
- Boot and Device Options
- USB Permissions
- Fan Considerations
- Boot Passwords and Secure Boot
- Trusted Platform Modules

Lesson 4.2

Power and Disk Issues

Troubleshoot Power Issues (Slide 1 of 2)

Follow
power flow

- Outlet to power cable
- Power cable to power supply
- Power supply to motherboard

Verify each
component
individually

- Power supply tester

Troubleshoot Power Issues (Slide 2 of 2)



Image by Konstantin Malkov @123RF.com

Troubleshoot POST Issues

Verify any changes

Check cables and connections

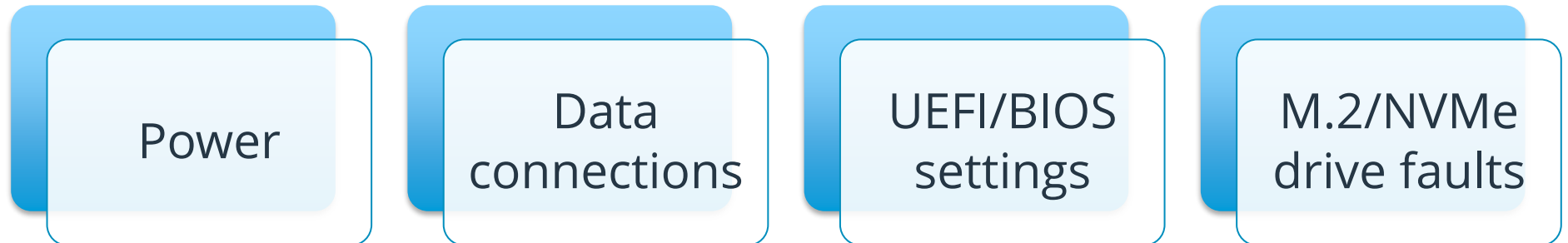
Faulty interfaces

Verify PSU

Faulty CPU or firmware issues

Beep codes

Troubleshoot Boot Issues



Power

Data
connections

UEFI/BIOS
settings

M.2/NVMe
drive faults

Troubleshoot Boot Sector Issues

MBR/GPT errors

Display connections

Boot error

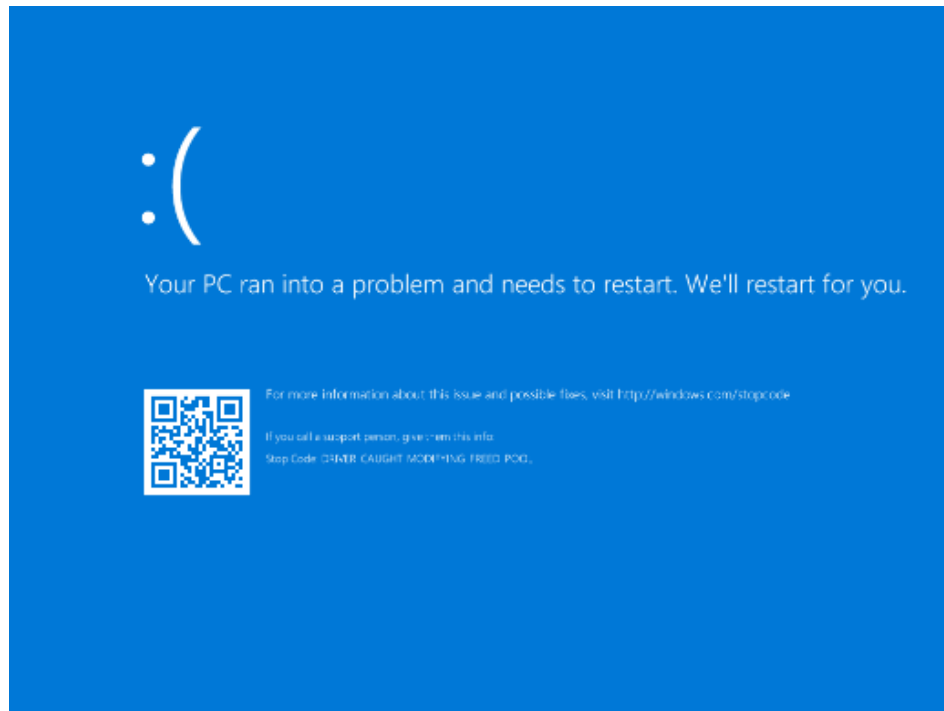
- Verify beep codes

Malware issues

Repair options

Troubleshoot OS Errors and Crash Screens

- Blue screen of death - BSOD
 - Device driver issues
 - Corrupt system files
 - Defective hardware
 - Overheating or PSU issues



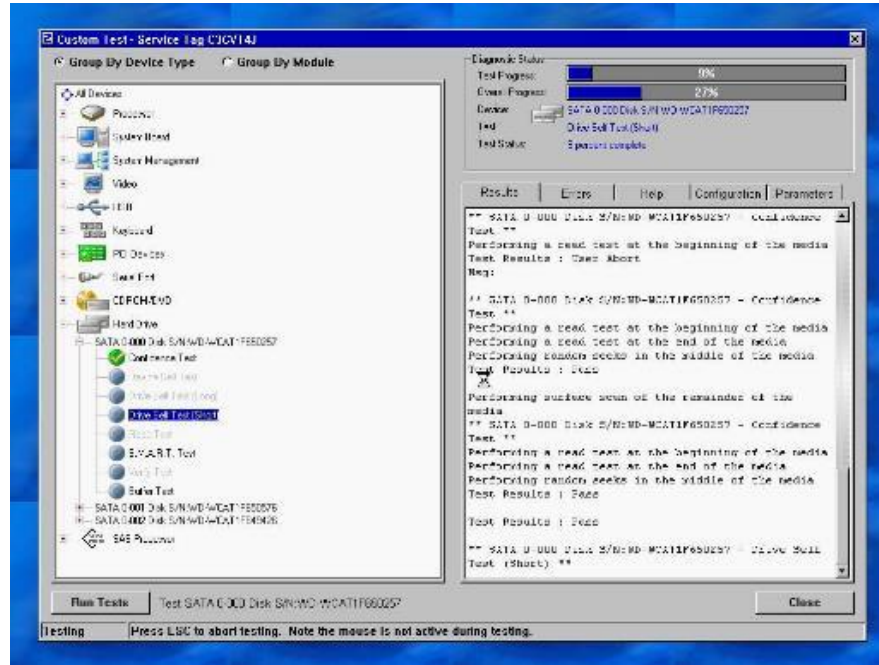
Screenshot courtesy of Microsoft.

Troubleshoot Drive Availability

- Unusual noise
- LED status/activity lights
- Boot device not found
- Missing drive in OS
- Read/write failure
- Audible alarms
- BSOD

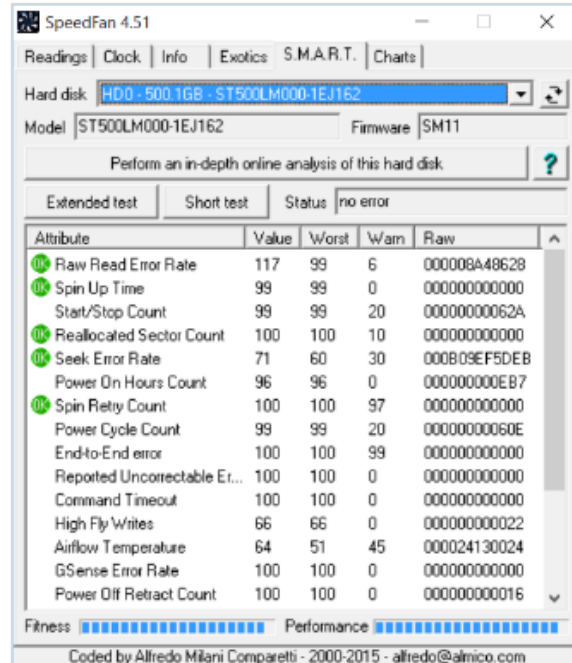
Troubleshoot Drive Reliability and Performance (Slide 1 of 3)

- S.M.A.R.T. drive tests



Screenshot courtesy of Microsoft.

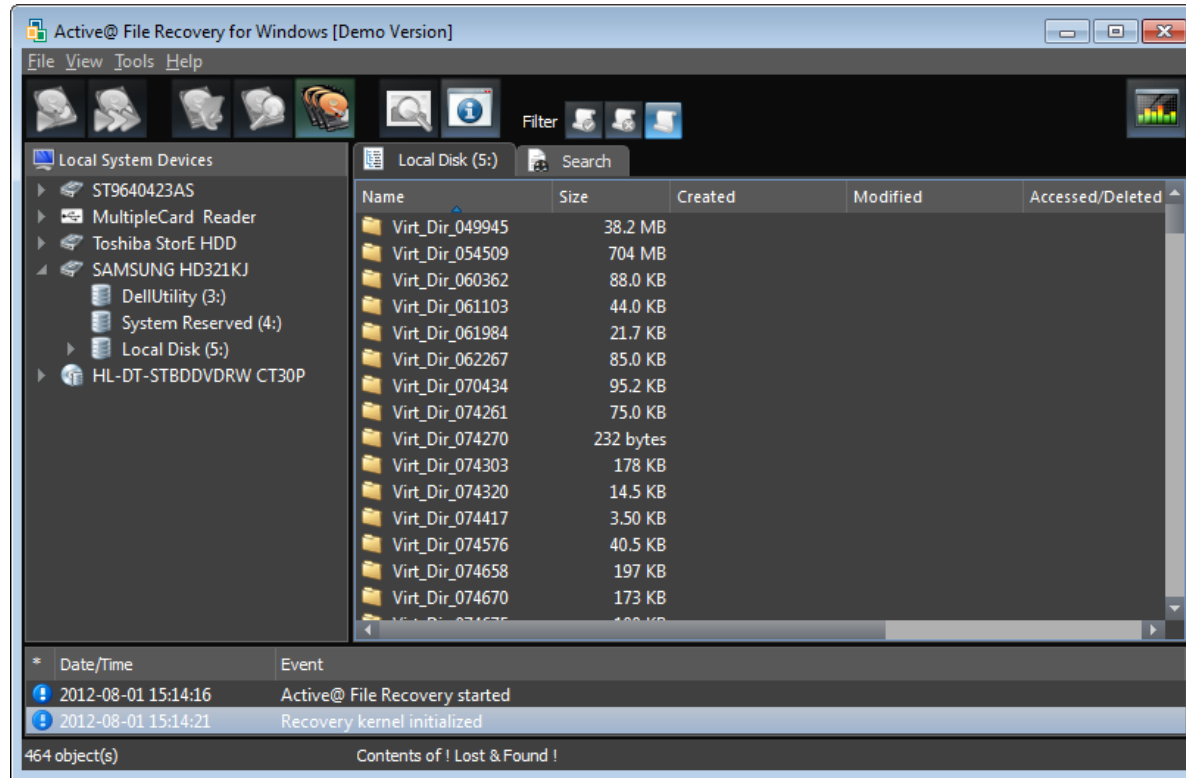
Troubleshoot Drive Reliability and Performance (Slide 2 of 3)



Screenshot courtesy of Microsoft.

- Input/Output Operations Per Second (IOPS)
- Bad sectors

Troubleshoot Drive Reliability and Performance (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Troubleshoot RAID Failure

Device failure
Array failure
Controller alerts

```
LSI Corp Config Utility  For Dell PERC H200  v7.01.09.00 (2010.03.22)
View Volume -- SAS2008
Volume                      1 of 1
Identifier
Type                         RAID 1
Size(GB)                     232
Status                       Inactive

Manage Volume

Slot  Device Identifier      RAID  Hot  Drive  Pred  Size
Num   ATA      WDC WD2502ABYS-13B05  Yes  No   Status Fail (GB)
1     ATA      WDC WD2502ABYS-13B05  Yes  No   Inactive No  232
---   ---      ---                      Yes  No   Missing ---  -----

Esc = Exit Menu      F1/Shift+1 = Help
Enter=Select Item   Alt+N=Next Volume
```


Troubleshoot RAID Failure

```
F10 = System Services
F11 = BIOS Boot Manager
F12 = PXE Boot

One 2.40 GHz Quad-core Processor, Bus Speed:4.80 GT/s, L2/L3 Cache:1 MB/8 MB

System Memory Size: 4.0 GB, System Memory Speed: 1067 MHz


Broadcom NetXtreme II Ethernet Boot Agent v5.0.5
Copyright (C) 2000-2009 Broadcom Corporation
All rights reserved.
Press Ctrl-S to Configure Device (MAC Address - 842B2B19E291)


Dell PERC H200/6Gbps SAS HBA BIOS
MPT2BIOS-7.01.09.00 (2010.03.22)
Copyright 2000-2009 LSI Corporation.


Integrated RAID exception detected:
  Volume (Hd1:079) is currently in state INACTIVE/OPTIMAL
Enter the Dell PERC H200/HBA Configuration Utility to investigate!

Press Ctrl-C to start Dell PERC H200/HBA Configuration Utility..
```

Lab Activity

- Lab: Troubleshoot System Power
(available in CertMaster Learn and CertMaster Perform)
- Lab: Troubleshoot System Startup
(available in CertMaster Learn and CertMaster Perform)
- Lab: Troubleshoot Boot Issues
(available in CertMaster Learn and CertMaster Perform)
- Lab: Troubleshoot Drive Availability
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Power and Disk Issues

- Power Issues
- POST Issues
- Boot Issues
- Boot Sector Issues
- OS Errors and Crash Screens
- Drive Availability
- Drive Reliability and Performance
- Troubleshoot RAID Failure

Lesson 4.3

System and Display Issues

Troubleshoot Component Issues

- Diagnostic steps
 1. Eliminate software issues
 2. Identify patterns
 3. Check power supply
 4. Suspect hardware
 5. Observe physical symptoms

Overheating

Temperature sensors

CPU and case fans

Heat sink

Blanking plates

Dust and ambient heat

Environmental issues

Physical Damage

ESD, power spikes

Overheating

Careless insertion of connectors

Dirt

Liquid spills

Scorch marks

Swollen capacitors



Lesson Review: Troubleshoot Performance Issues

Overheating

Misconfigurations

Verify the problem

Rule out software
and networking
issues

Troubleshoot Inaccurate System Date/Time (Slide 2 of 2)



Real-Time Clock (RTC) or CMOS battery (legacy)

Provides power to maintain settings and configuration

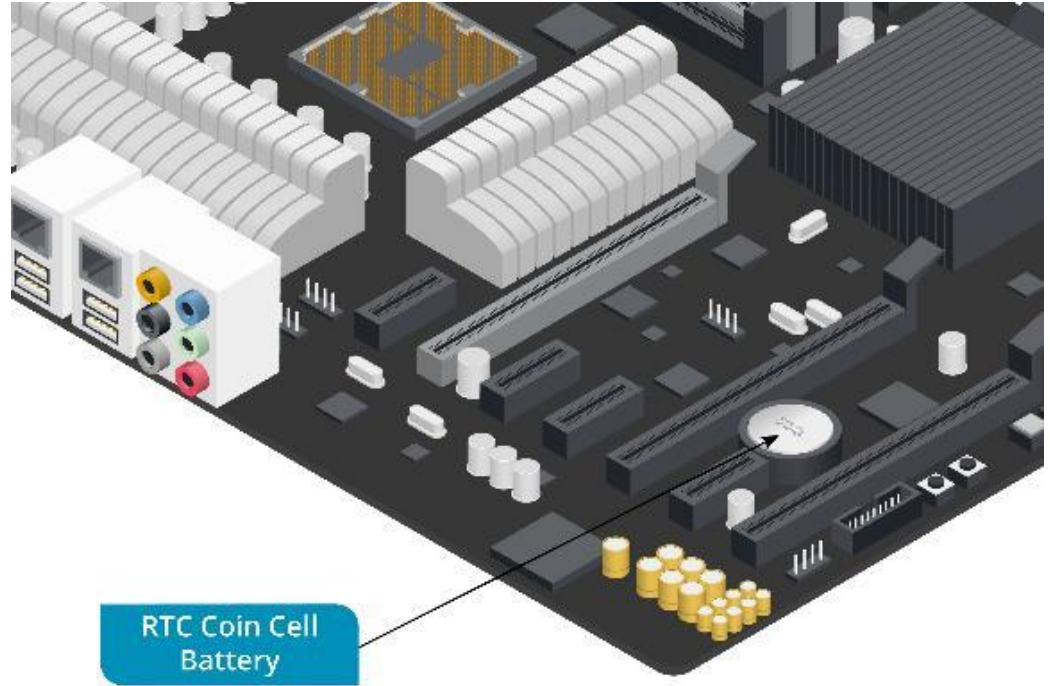


Image ©123RF.com

Troubleshoot Missing Video Issues



Physical cable issues



Burnt-out bulbs



Projector
shutdown

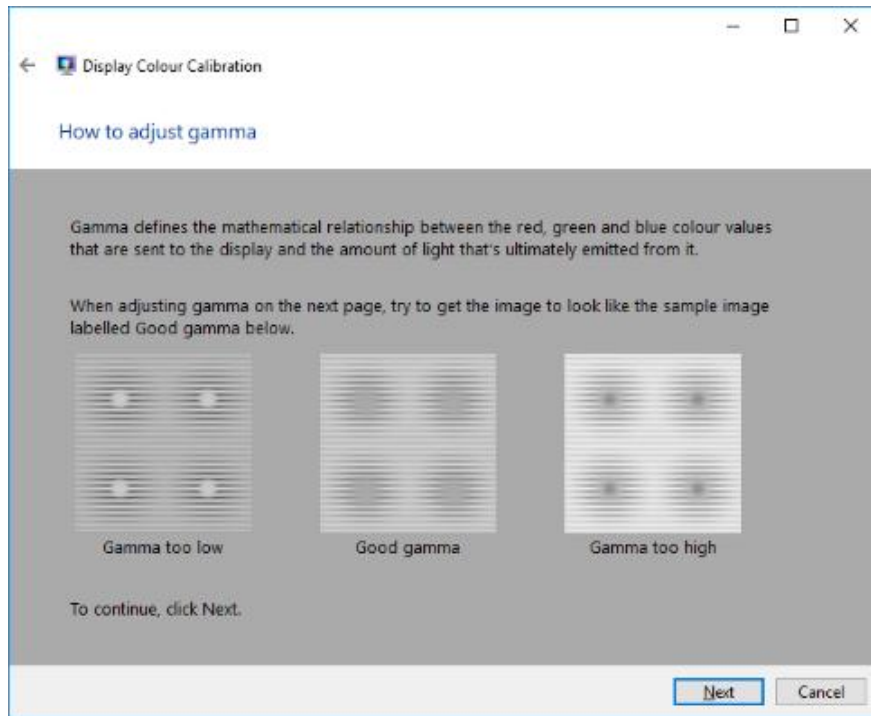
Overheating
Blown bulb



- Image ©123RF.com

Troubleshoot Video Quality Issues

- Dim/fuzzy images
- Flashing screen
- Dead pixels
- Burn-in
- Incorrect color



- Screenshot courtesy of Microsoft.

Lab Activity

Lab: Troubleshoot GPU
(available in CertMaster Learn and CertMaster Perform)



Lab: Resolve PC Hardware Support Tickets
(available in CertMaster Learn and CertMaster Perform)



Lab: Troubleshoot a Malfunctioning Computer
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: System and Display Issues

- Troubleshoot Component Issues
- Overheating
- Physical Damage
- Performance Issues
- Inaccurate System Date/Time
- Missing Video Issues
- Video Quality Issues